T2T GAPLESS GENOME ASSEMBLY OF THE NEGLECTED CROP *MORINGA STENOPETALA*

A. Gálvez-Salido, C. Martínez Padilla, P. Contreras Saavedra, A. Ríos Bravo and L. Carretero-Paulet*

Department of Biology and Geology, “Pabellón de Historia Natural-Centro de Investigación de Colecciones Científicas de la Universidad de Almería” (PHN-CECOUAL), University of Almería, E04120 Almería (Spain), ags408@ual.es (**presenting author**); lpaulet@ual.es (*corresponding author)

Introduction

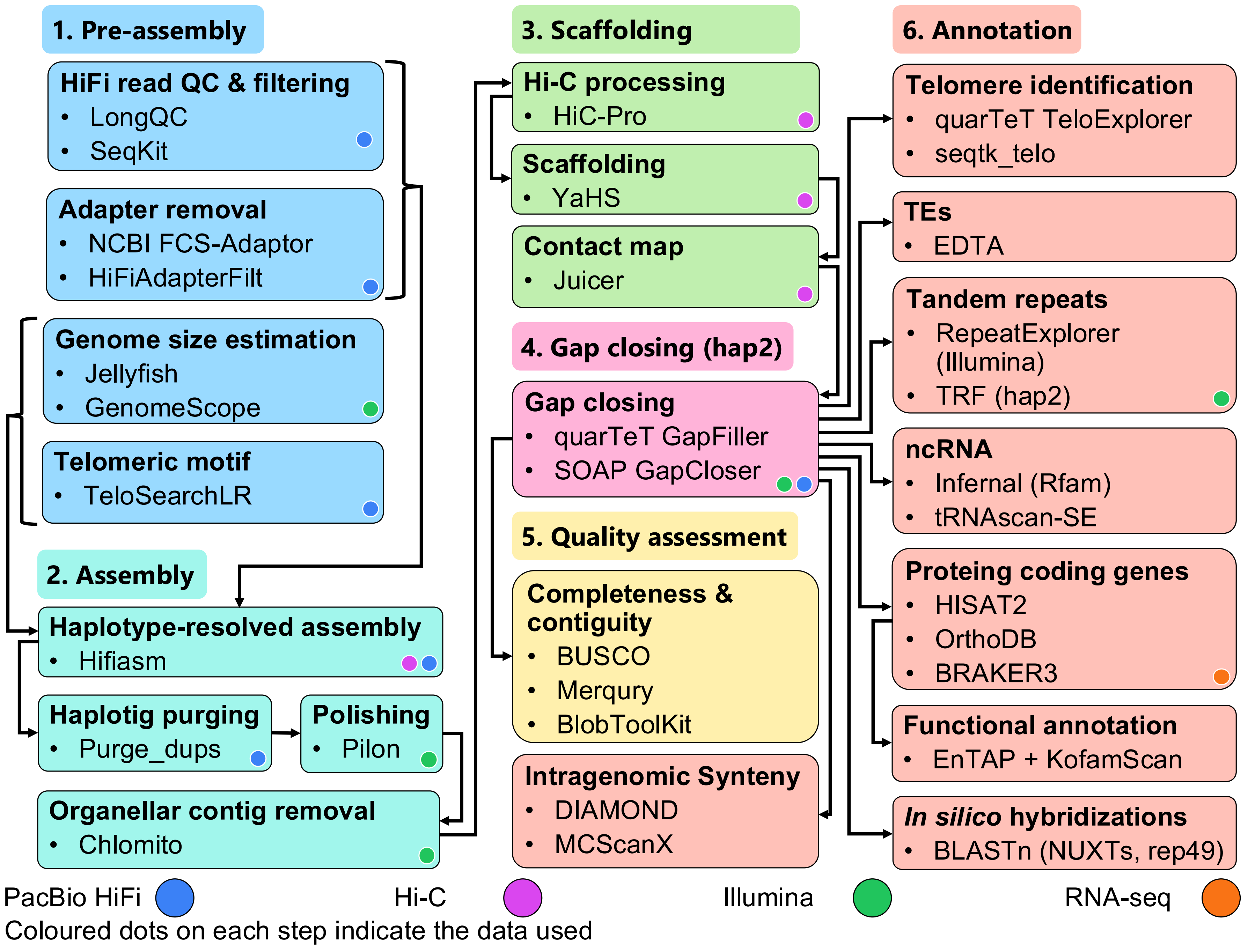
- ***Moringa stenopetala*** is the second most cultivated species of Moringa, the sole genus of the Moringaceae family. This family exhibits rapid phenotypic diversification in morphology and biochemistry, including **pharmacologically relevant glucosinolates** (Olson, 2002; Fahey et al., 2018).
- **Native to Ethiopia and Kenya**, *M. stenopetala* is a key source of food, fodder, medicine, oil, and natural water flocculants. It is also remarkably drought tolerant (Abiyu et al., 2018; Galgaye et al., 2020). However, it remains underutilised and understudied, as well as being **semi-domesticated**.
- The Moringaceae family is also a model for **organelle-to-nucleus DNA transfer**. *M. oleifera* has the highest nuclear plastid DNA (NUPT) content reported in plants (3.29% versus ~0.1%), which was fixed after recent burst events (Marczuk-Rojas et al., 2024). **Whether this is a genus-wide feature remains unknown** (Carretero-Paulet et al., 2026).

OBJECTIVE

Generate the first telomere-to-telomere (T2T) genome assembly of *M. stenopetala*:

- Annotate repetitive elements.
- Annotate non-coding RNA genes.
- Annotate protein-coding genes.
- Perform the functional annotation.

Materials and methods



1. Pre-assembly

HiFi read QC & filtering
- LongQC
- SeqKit ●

Adapter removal
- NCBI FCS-Adaptor
- HiFiAdapterFilt ●

Genome size estimation
- Jellyfish
- GenomeScope ●

Telomeric motif
- TeloSearchLR ●

2. Assembly

Haplotype-resolved assembly
- Hifiasm ●●

Haplotig purging
- Purge_dups ●

Polishing
- Pilon ●

Organellar contig removal
- Chlomito ●

3. Scaffolding

Hi-C processing
- HiC-Pro ●

Scaffolding
- YaHS ●

Contact map
- Juicer ●

4. Gap closing (hap2)

Gap closing
- quarTeT GapFiller
- SOAP GapCloser ●●

5. Quality assessment

Completeness & contiguity
- BUSCO
- Merqury
- BlobToolKit

Intragenomic Synteny
- DIAMOND
- MCScanX

6. Annotation

Telomere identification
- quarTeT TeloExplorer
- seqtk_telo

TEs
- EDTA

Tandem repeats
- RepeatExplorer (Illumina)
- TRF (hap2) ●

ncRNA
- Infernal (Rfam)
- tRNAscan-SE

Proteing coding genes
- HISAT2
- OrthoDB
- BRAKER3 ●

Functional annotation
- EnTAP + KofamScan

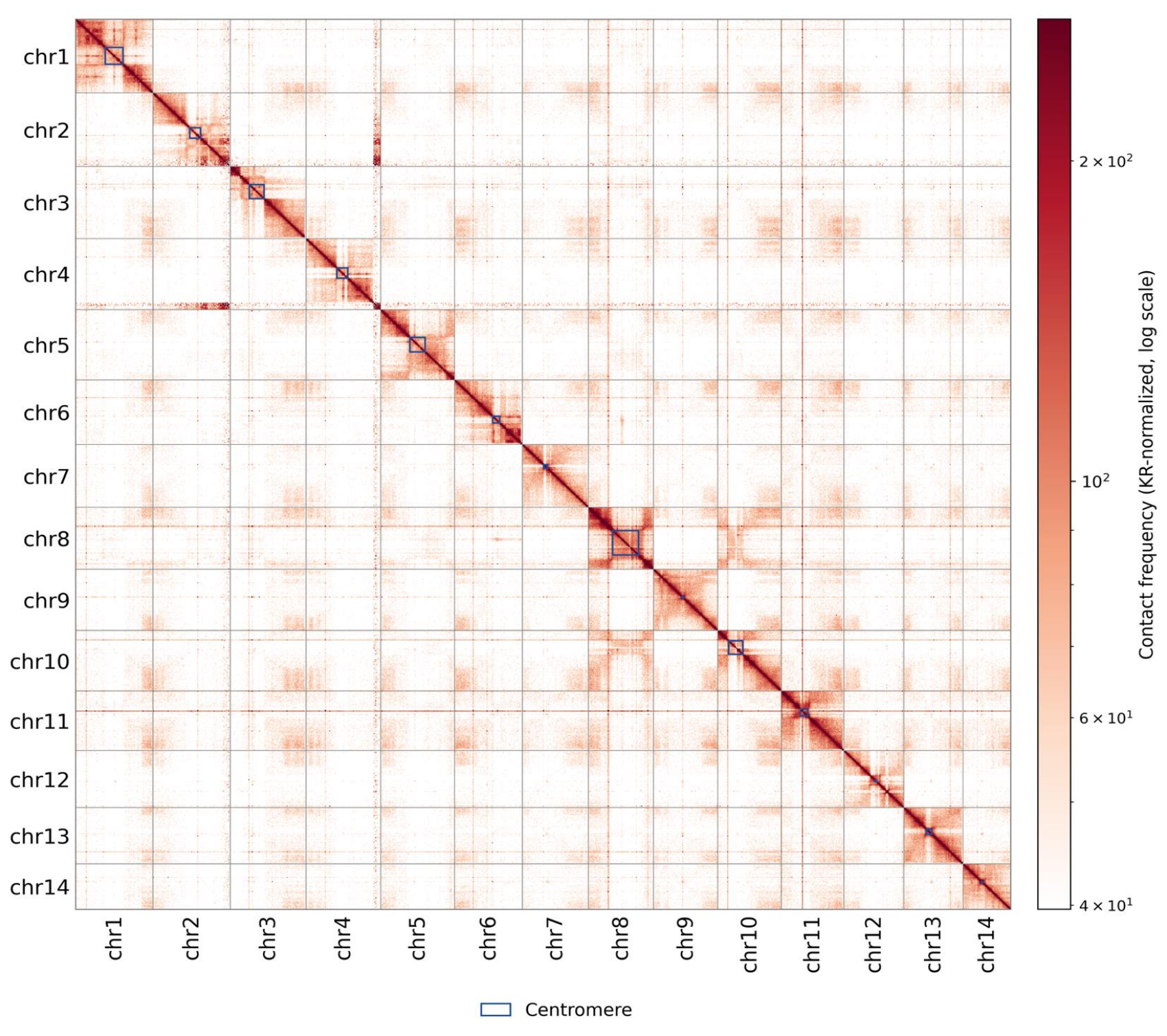
***In silico* hybridizations**
- BLASTn (NUXTs, rep49)

PacBio HiFi ● Hi-C ● Illumina ● RNA-seq ●

Coloured dots on each step indicate the data used

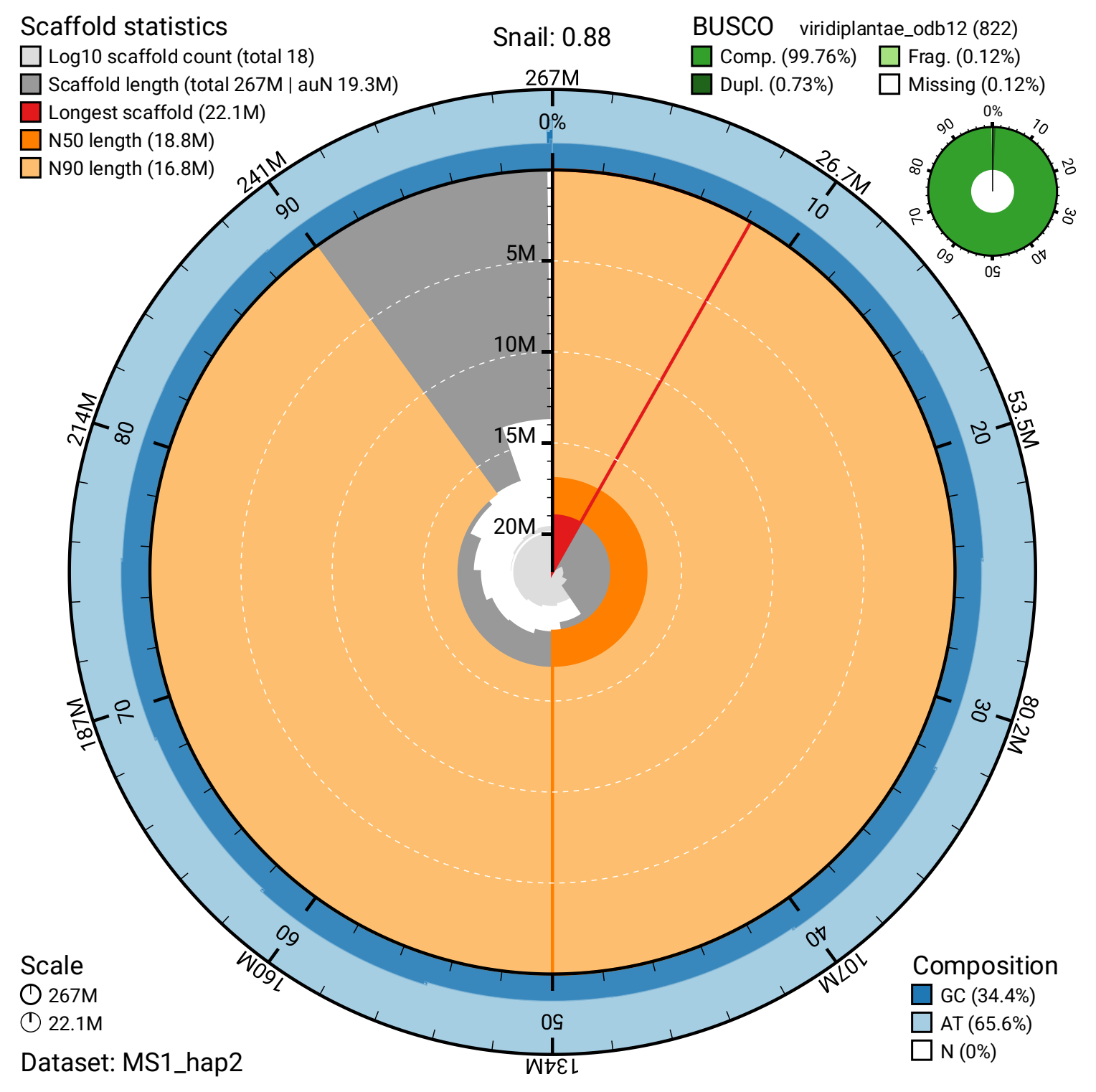
Results

Hi-C contact map of the hap2 assembly



- The genome is anchored into 14 chromosomes.
- Strong signal along the diagonal.
- No sign of large misassemblies.

Snail plot of the hap2 assembly



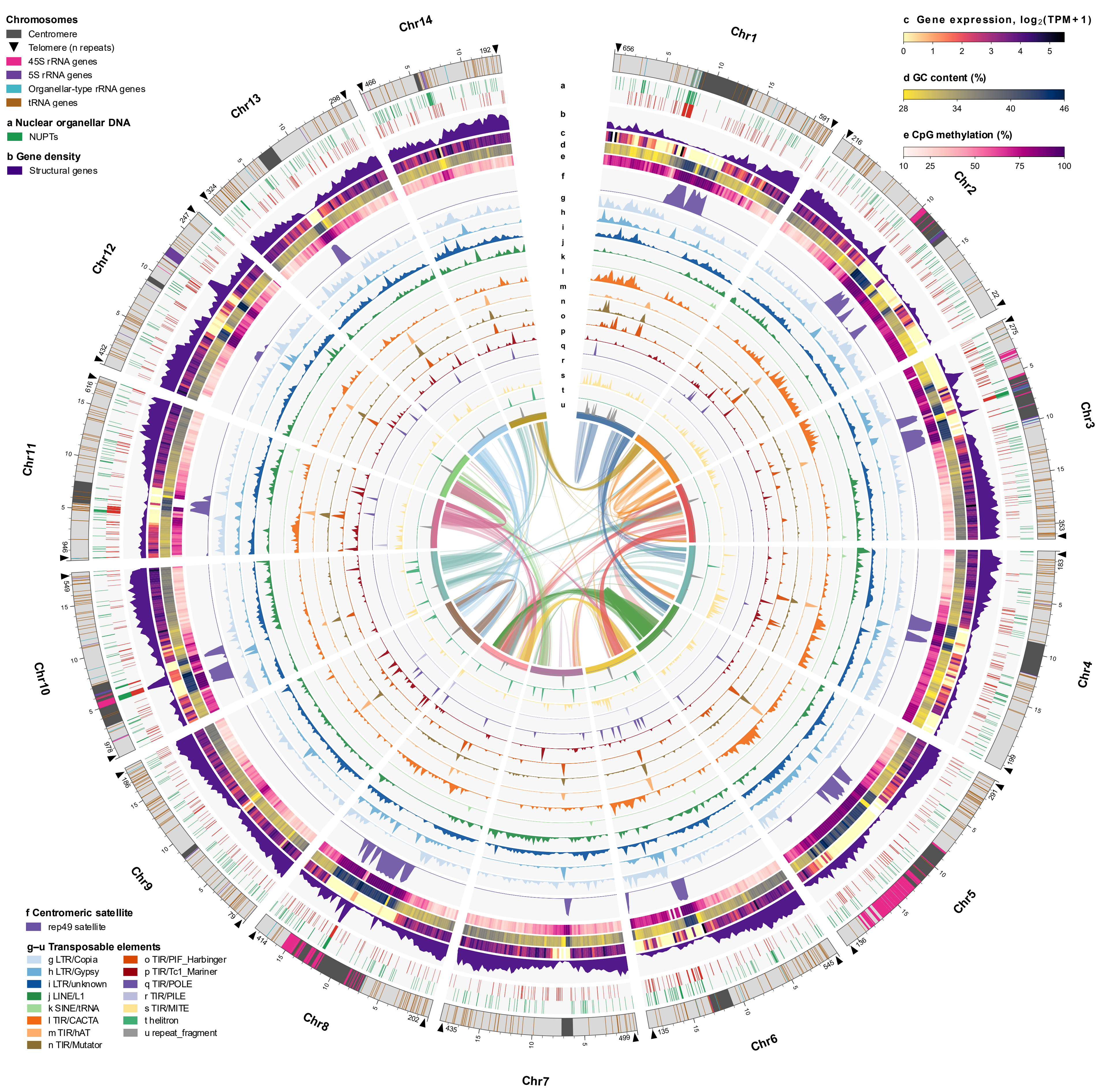
- 14 chromosomes and 4 unplaced scaffolds.
- Truly gapless, no Ns.
- Very high gene-space completeness and low redundancy.

Functional annotation

Annotation type	Annotated genes	% of genes	Total annotations	Mean per annotated gene	Median per annotated gene	Max per gene	Distinct terms
Similarity search							
Best-hit accessions	19,747	81.80	20,519	1.04	1	5	18,711
eggNOG-mapper							
Orthologous groups	19,998	82.84	90,600	4.53	5	12	53,902
COG categories	18,241	75.56	19,229	1.05	1	5	24
Pfam domains	17,645	73.09	29,110	1.65	1	9	4,157
EC numbers	4,106	17.01	4,860	1.18	1	8	1,293
CAZy families	437	1.81	501	1.15	1	3	70
InterProScan							
Signatures	19,412	80.41	121,087	6.24	6	35	20,623
InterPro entries	18,229	75.51	66,648	3.66	3	20	11,286
Pfam	17,400	72.08	27,040	1.55	1	10	5,415
PANTHER families	18,955	78.52	18,982	1.00	1	3	6,245
Gene Ontology							
GO terms, eggNOG-mapper	10,554	43.72	434,703	41.19	34	622	9,954
GO terms, InterProScan	9,718	40.26	24,128	2.48	2	13	3,081
GO terms, UniProt	5,601	23.20	36,969	6.60	6	53	4,292
GO terms, union of sources	14,534	60.21	465,257	32.01	24	629	10,754
GO non-redundant, BP	12,413	51.42	130,232	10.49	6	274	5,979
GO non-redundant, MF	11,552	47.85	40,008	3.46	2	29	2,560
GO non-redundant, CC	12,195	50.52	41,520	3.40	3	27	1,010
GO non-redundant, all aspects	14,534	60.21	211,673	14.56	10	307	9,549
GO propagated, all aspects	14,534	60.21	559,374	38.49	31	643	10,975
KEGG							
KO (final)	11,515	47.70	12,364	1.07	1	6	4,782
KO KofamScan	9,613	39.82	10,013	1.04	1	3	4,353
KO eggNOG-mapper	9,478	39.26	10,421	1.10	1	6	4,061
Plant pathways	5,735	23.76	13,638	2.38	2	16	151
Modules	2,366	9.80	3,574	1.51	1	11	281

- 81.8% of the genes have a similarity hit, and 80.4% carry at least one InterProScan signature.
- 60.2% were assigned GO terms, giving 10,754 distinct terms across all sources.
- 47.7% received a KEGG orthology (KO) assignment, which maps them to 151 plant pathways and 281 modules.

Circos plot showing the complete genome annotation



Conclusion

We deliver the first **gapless T2T genome** of *M. stenopetala*: 14 chromosomes, **fully annotated**, turning a neglected crop into a genomic resource.

Acknowledgements

This research was supported by the projects: “El modelo agrícola de Almería ante el cambio global. Propuestas desde la genómica de la agrobiodiversidad (OrphanEvolGen)” (PID2023-146207OB-I00), funded by the 2023 Call for Knowledge Generation Projects and associated predoctoral training actions of the Spanish Ministry of Science and Innovation, and “Puesta en valor del cultivo huérfano etíope *Moringa stenopetala* mediante la secuenciación y anotación funcional de su genoma (shiferawgenome)” (UAL2024_1) funded by the 2024 Plan Propio for International Cooperation (Program I) of the University of Almería, granted to Dr. Lorenzo Carretero-Paulet.

References

UNIVERSIDAD DE ALMERÍA
ciencias experimentales
PABELLÓN DE HISTORIA NATURAL
PHN
cecoual
Centro de Colecciones Científicas de la Universidad de Almería
GOBIERNO DE ESPAÑA
MINISTERIO DE CIENCIA, INNOVACIÓN Y UNIVERSIDADES
XV SIMPOSIO de Investigación EN CIENCIAS EXPERIMENTALES
SCAN TO LEARN MORE ABOUT ME!